Narration Script

MAP-A rationale video v6 · MCS and the SDGs (1:45) · version without narration

Length 1:45 · 1920 × 1080 · 30 fps · subtitles burned in · 248 words in four scenes, then a 7.5-second title card with no narration. Read each line over **MAP-A_Rationale_Video_v6_MCS_SDG_No_Narration.mp4**. Each line starts when its subtitle appears, and every visual cue is tied to a word, so match your pace to the subtitles.

How to record	Say it this way
1. Play the video and start speaking when each subtitle appears (**Start**). 2. Finish the line by **Finish by**. **Read time** is the length of the reference read in the narrated version. 3. Record each scene as its own take. Scene windows: 0:00–0:17.2, 0:17.2–0:35.0, 0:35.0–0:59.4, 0:59.4–1:37.5, then the title card 1:37.5–1:45 (no narration). 4. The boom lands at 1:23.0. Read the conclusion (4.7 to 4.10) after it, with a full pause before "Distance…", then stay silent through the title card to 1:45. 5. Pace is about 2.2 words per second, with a short breath at each comma.	**MCS**: "M-C-S", letter by letter **Dijkstra's**: DIKE-struhz **MAP-A**: "Map A" **Albay**: al-BYE **jeepney**: JEEP-nee **Goals 3, 10, and 11**: "Goals three, ten, and eleven" **e-mail**: EE-mail **"test network"**, never "Albay", for the a3, S3 and S4 numbers **RA 10932**, ₱ fares and table values are shown on screen only. Do not read them.

Scene 1 · Trip

0:00.0–0:17.2 · 41 words · bar: Convenience

#	Start	Finish by	Read time	Narration (read this) · on screen
1.1	0:00.3	0:03.1	2.7 s	**Start with one trip, from point A to point B.** On screen: Albay map draws in; pins A and B drop; a dotted path with a "?" joins them.
1.2	0:03.5	0:07.0	3.4 s	**Which jeepney do you ride, when do you arrive, and what do you pay?** On screen: Three tags pop in, one per question: Which jeepney? · When do you arrive? · What do you pay?
1.3	0:07.6	0:12.5	4.8 s	**In Albay, no machine-readable record of routes, stops, or fares can tell you.** On screen: The map dims while unrecorded vehicles drive between towns. On "no machine-readable record" the tags turn grey; the notation "R = S = H = F = empty set" (routes · stops · headways · fares); chips Jeepney · Tricycle · Minibus · UV van and "No routes, stops, headways, or fares on record", with "fares" in gold.
1.4	0:13.0	0:14.7	1.6 s	**Our timetable model can.** On screen: Synthetic test network with "Timetable graph G = (V, E)"; gold route a3 → S4; card "Door to door: 18.6 min" with τ(a3, S4) = 18.6 min; the fare meter counts to ₱20.00 for 7 km on a traditional jeepney, then "₱16.00 with 20% discount" and f(d) = 14 + 2·max(0, d − 4). Hold to 0:17.2.

Scene 2 · Community

0:17.2–0:35.0 · 47 words · bar: Fairness

#	Start	Finish by	Read time	Narration (read this) · on screen
2.1	0:17.4	0:18.3	0.8 s	**Zoom out.** On screen: The camera pulls back from the route to Albay: town boundaries, main roads and the four urban centres.
2.2	0:18.7	0:25.7	6.9 s	**The same model counts the schools, markets, and hospitals a person can reach in thirty minutes, and who is left out.** On screen: Schools, markets and hospitals appear clustered around the town centres; $A_i(30) = \Sigma\ 1[\tau(i, j) \le 30\ \text{min}]$; a road-shaped 30-minute reach area draws around Legazpi City; facilities inside light up, those outside grey out; “$L = \{ i : A_i(30) = 0 \}$” on “who is left out”.
2.3	0:26.0	0:31.8	5.8 s	**Sustainable Development Goals 3, 10, and 11 ask for health, equality, and transport for all.** On screen: The map dims under the header “SUSTAINABLE DEVELOPMENT GOALS THIS RESEARCH SUPPORTS”. On each number a tile lands: SDG 3 Good Health and Well-being (Target 3.8: right hospital, right bed; cost never decides an emergency) · SDG 10 Reduced Inequalities (Targets 10.2, 10.3: measures who is left out with Gini, Theil and Palma) · SDG 11 Sustainable Cities and Communities (Target 11.2: transport reach within 30 minutes, for all).
2.4	0:32.1	0:34.9	2.7 s	**In Albay, there was nothing to measure that against.** On screen: The three SDG tiles shrink into small badges at the bottom left, where they stay for the rest of the video. A “?” appears on the 30-minute reach area.

Scene 3 · Patient

0:35.0–0:59.4 · 59 words · bar: A life

#	Start	Finish by	Read time	Narration (read this) · on screen
3.1	0:35.3	0:36.7	1.3 s	**Now, an emergency.** On screen: The Albay map zooms back into the test network; the patient appears at a3 with a red pulse.
3.2	0:37.2	0:41.1	3.8 s	**The nearest hospital may be full, or lack the right level of care.** On screen: A red dashed line marks the nearest hospital, S3 (7.0 min), with h_dist = argmin d(a3, h) = S3. On "full" S3 turns red: b(S3) = 0; card S3 · FULL · Level 1 · 0 free beds · 7.0 min.
3.3	0:41.7	0:46.2	4.5 s	**Maps show where hospitals are, not free beds or what a family can afford.** On screen: Card "A map shows where. It does not show free beds, level, or cost." S4 shows "free beds ? · level ? · cost ?".
3.4	0:47.0	0:54.9	7.8 s	**Our hospital mode collects free-bed reports by e-mail** [0:50.8] **and sends the ambulance to the nearest hospital that can take the patient.** On screen: "HOSPITAL MODE"; six e-mails fly to S4 and the counter reaches 6/6; S4 turns gold, b(S4) = 8 (Level 3 · 18.6 min); the decision rule $h^* = \text{argmin } \tau(a3, h)$ subject to level of care, open ER, free beds and budget or emergency acuity; the ambulance carries the patient past S3; $h^* = S4$; check chips.
3.5	0:55.6	0:58.7	3.0 s	**In an emergency, payment never delays the route.** On screen: Chip "Payment waived for acuity 1 and 2 (RA 10932)" with "k in K_E = {1, 2}". After the line, hold: "Distance-only routing sent 5 of 6 anchors to a full hospital".

Scene 4 · How MCS comes into play

0:59.4–1:37.5 · 101 words · bar: MCS

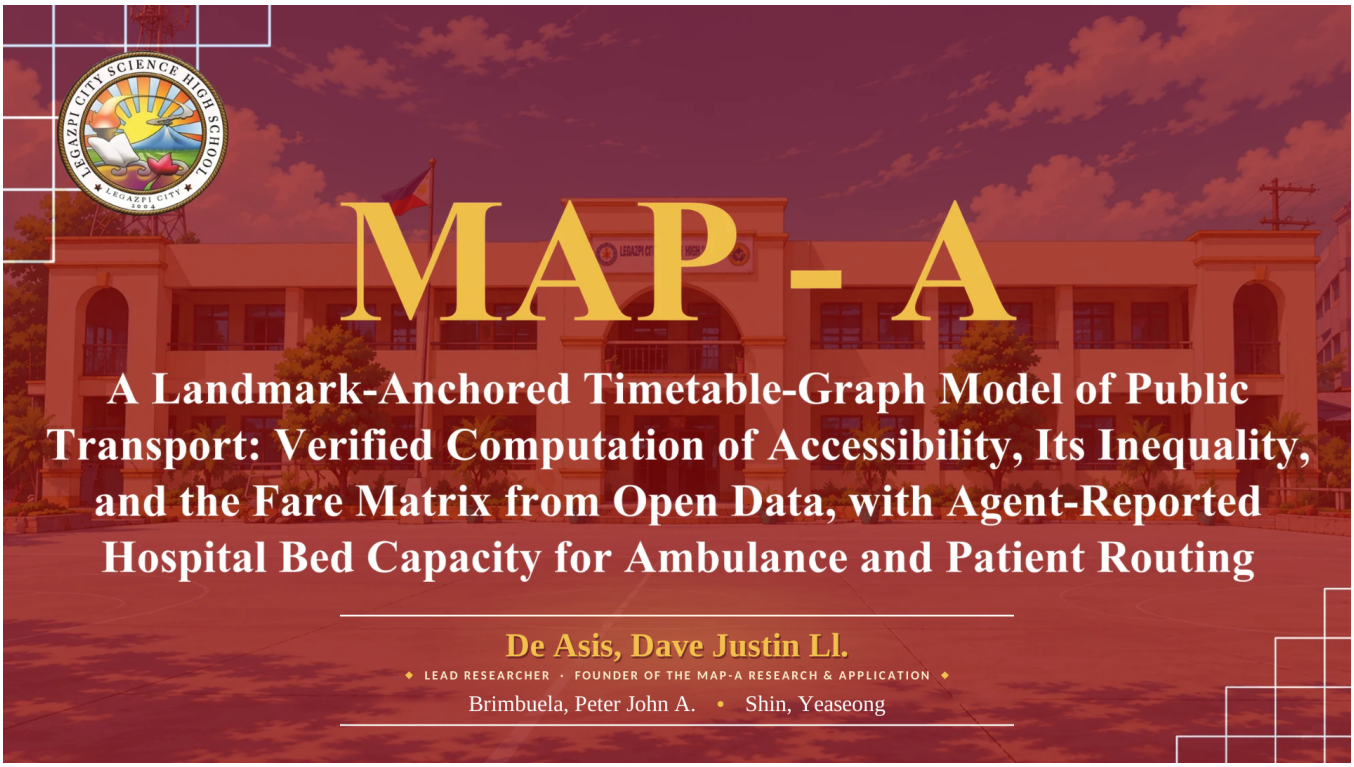
#	Start	Finish by	Read time	Narration (read this) · on screen
4.1	0:59.5	1:02.5	2.9 s	**You may be wondering: how does MCS come into play?** On screen: Scene title. "You may be wondering…", then "How does". On "MCS" the gold letters M, C, S drop in and unfold into Mathematical · Computational · Sciences, then "come into play?". Faint math symbols drift behind; the stakes bar climbs to its top rung, "MCS".
4.2	1:02.9	1:05.4	2.5 s	**Mathematics turns open data into a graph.** On screen: The title shrinks into an "MCS" badge and a chain lights up across the top: Open data → Graph → Algorithm → Numbers → Optimisation → Proof. The synthetic test network builds stop by stop (a3, S3, S4). Panel "MATHEMATICS": G = (V, E); V = stops and landmarks, E = ride, wait, transfer and walk arcs; built from open maps, registries, census counts and fares.

#	Start	Finish by	Read time	Narration (read this) · on screen
4.3	1:05.8	1:10.2	4.3 s	**Computing searches it with Dijkstra's algorithm for the exact fastest trip.** On screen: Panel “COMPUTING”: Dijkstra’s label-setting search, $\tau^*(o, d, t0) = t0 + \min$ over paths of $\Sigma\ \omega(u, v)$, with $\omega(u, v) = t(v) - t(u) \geq 0$. On “Dijkstra’s” a wave spreads from a3 and every stop settles with its travel time in minutes (S3 7.0 min, S4 18.6 min). On “fastest” the route a3 → S4 lights gold; card “Proposition 1 · returns the exact earliest arrival” and $\tau(a3, S4) = 18.6$ min.
4.4	1:10.7	1:14.8	4.0 s	**Those times count access, measure inequality, and price every fare.** On screen: On “access”: stops within 30 minutes of a3 glow and the rest grey out; panel $A_i(30) = \Sigma\ 1[\tau(i, j) \leq 30\ \text{min}]$. On “inequality”: a Lorenz curve and the population-weighted Gini coefficient $G = (1/2\mu)\ \Sigma\ \Sigma\ p_a\ p_b\ |A_a - A_b|$. On “fare”: a fare matrix fills cell by cell, $F_{ij} = f(d_{ij})$ with $f(d) = 14 + 2 \cdot \max(0, d - 4)$; the 7 km pair shows ₱20.
4.5	1:15.2	1:19.8	4.6 s	**For a patient, it finds the fastest hospital with the right care and a free bed.** On screen: Panel “OPTIMISATION”: $h^* = \text{argmin}\ \tau(a3, h)$ subject to $\ell(h) \geq \ell_k$ (right level of care), ER(h) = 1 (ER open), b(h) > 0 (a free bed), c(h) ≤ B or k K_E (within budget, or an emergency). On “patient” S3 turns red: FULL · 0 free beds, “S3 is nearest, but b(S3) = 0”. The ambulance drives a3 → S4, each condition gets a check, and h^* = S4.
4.6	1:20.3	1:22.9	2.5 s	**Twelve results, proven, then verified.** On screen: “12 results = 9 propositions + 3 corollaries”: twelve tiles (Props. 1, 2, 3, 5, 6, 7, 9, 10, 12; Cors. 4, 8, 11) get a proof mark on “proven” and a check on “then verified”, with “Each proven on paper, then checked against independent code on synthetic instances · no field data”. They implode into a glowing core as a riser builds.
4.7	1:23.5	1:25.5	2.1 s	**MAP-A will not add one bed to Albay.** On screen: Conclusion. A boom at 1:23.0 clears the screen to a plain background. The line appears large, with “one bed” in gold. The large text is its subtitle.
4.8	1:26.0	1:30.9	5.0 s	**But once hospitals report, it will tell him which one has a free bed and the right care.** On screen: Appears under it in two rows, with “free bed” and “right care” in gold. Both lines then fade out.
4.9	1:32.0	1:34.2	2.3 s	[pause] **Distance tells you where the hospital is.** On screen: [pause] The screen is empty for a moment. Then the line appears, with the MAP-A logo and title below.
4.10	1:34.6	1:37.2	2.6 s	**Access tells you whether it can help in time.** On screen: Lands in large gold type with a final hit; “in time” is underlined. Hold silence. At 1:37.5 the title card follows.

Title card · Researchers

1:37.5–1:45.0 · no narration

#	Start	Finish by	Read time	Narration (read this) · on screen
5.1	1:37.5	1:45	—	**[No narration. Stay silent; let the music and the title card play.]** On screen: 1:37.5 The conclusion frame crossfades into the title slide: school building under a red overlay, Legazpi City Science High School seal, white corner grid. 1:39.1 The gold "MAP - A" drops in letter by letter and lands with a hit; a gold shine sweeps across it. 1:39.4 The full title fades in line by line: "A Landmark-Anchored Timetable-Graph Model of Public Transport: Verified Computation of Accessibility, Its Inequality, and the Fare Matrix from Open Data, with Agent-Reported Hospital Bed Capacity for Ambulance and Patient Routing". 1:40.6 Two white lines draw outward. 1:41.1 **De Asis, Dave Justin Ll.** in large gold type; 1:41.6 under it "LEAD RESEARCHER · FOUNDER OF THE MAP-A RESEARCH & APPLICATION"; 1:42.1 below, "Brimbuela, Peter John A. · Shin, Yeaseong" in white. 1:42.8 A gold shine and glow pass over the leader's name. 1:44.3 Fade to black; the video ends at 1:45.



Title card, final frame (also supplied as MAP-A_Title_Page_1920x1080.png and MAP-A_Title_Page_3840x2160.png for the title slide).

Clean read

The same narration without the production notes. Use it as a teleprompter or to rehearse. A slash marks a short breath.

SCENE 1 · TRIP · 0:00.0–0:17.2

0:00.3 Start with one trip, / from point A to point B.

0:03.5 Which jeepney do you ride, / when do you arrive, / and what do you pay?

0:07.6 In Albay, / no machine-readable record of routes, / stops, / or fares can tell you.

0:13.0 Our timetable model can.

SCENE 2 · COMMUNITY · 0:17.2–0:35.0

0:17.4 Zoom out.

0:18.7 The same model counts the schools, / markets, / and hospitals a person can reach in thirty minutes, / and who is left out.

0:26.0 Sustainable Development Goals 3, / 10, / and 11 ask for health, / equality, / and transport for all.

0:32.1 In Albay, / there was nothing to measure that against.

SCENE 3 · PATIENT · 0:35.0–0:59.4

0:35.3 Now, / an emergency.

0:37.2 The nearest hospital may be full, / or lack the right level of care.

0:41.7 Maps show where hospitals are, / not free beds or what a family can afford.

0:47.0 Our hospital mode collects free-bed reports by e-mail and sends the ambulance to the nearest hospital that can take the patient.

0:55.6 In an emergency, / payment never delays the route.

SCENE 4 · HOW MCS COMES INTO PLAY · 0:59.4–1:37.5

0:59.5 You may be wondering: / how does MCS come into play?

1:02.9 Mathematics turns open data into a graph.

1:05.8 Computing searches it with Dijkstra's algorithm for the exact fastest trip.

1:10.7 Those times count access, / measure inequality, / and price every fare.

1:15.2 For a patient, / it finds the fastest hospital with the right care and a free bed.

1:20.3 Twelve results, / proven, / then verified.

1:23.5 **MAP-A will not add one bed to Albay.**

1:26.0 **But once hospitals report, it will tell him which one has a free bed and the right care.**

[pause]

1:32.0 **Distance tells you where the hospital is.**

1:34.6 **Access tells you whether it can help in time.**

TITLE CARD · RESEARCHERS · 1:37.5–1:45.0

1:37.5 [Silent. Title card: MAP - A · De Asis, Dave Justin Ll., Lead Researcher and Founder · Brimbuela, Peter John A. · Shin, Yeaseong. Ends 1:45.]